

Gustave Nganso

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SUMMARY

Lead engineer with nearly 20 years building production full-stack and backend systems, specializing in Python and Node.js services, LLM and RAG pipelines, and cloud-native delivery on AWS and Azure. Hands-on builder focused on API design, testing, and end-to-end ownership. Trilingual (English, French, German).

- Built and shipped a production RAG platform (Olotalk) on Qdrant, PostgreSQL, Redis, and ClickHouse, with Python ingestion workers and Claude-based enrichment.
- 6 years with Python and Django; 20 years across JavaScript/TypeScript and the React, Angular, and Vue ecosystems.
- Cloud-native delivery on AWS, Azure, and GCP using Docker and Kubernetes, with CI/CD on GitHub Actions and Jenkins.
- Pairs with AI coding tools day to day while keeping full review, debugging, and refactoring discipline.

TECHNICAL SKILLS

AI & LLM: LangChain, RAG, LLM orchestration, agentic workflows (2 yrs)

Python: Python, Django (6 yrs)

Backend: Node.js, Express, REST APIs, JSON, GraphQL; Golang (4 yrs), Rust (2 yrs)

Frontend: TypeScript, JavaScript/DOM (20 yrs), React/Next.js, Redux (7/4 yrs), Vue.js/Nuxt.js (3 yrs), Angular, AngularJS, NgRx, RxJS, Svelte, HTML/CSS

Cloud: AWS, Azure, GCP, Linode, DigitalOcean

Containers & CI/CD: Docker (6 yrs), Kubernetes, GitHub Actions, Jenkins, Git (10 yrs)

Databases: PostgreSQL, MySQL (5 yrs), MongoDB (8 yrs), Redis, ClickHouse, Qdrant

Practices: Agile SDLC, design patterns, automated unit and integration testing (Jest, Pytest, Mocha, Karma)

PROFESSIONAL EXPERIENCE

7971389 Canada Inc., Toronto

01/2009 – Present

Principal Full-Stack Developer & Instructor

- Run an independent software consultancy delivering full-stack and AI/LLM projects for fintech, media, and public-sector clients (engagements below).
- Teach full-stack and systems engineering in JavaScript, TypeScript, Python, Golang, and Rust, and mentor junior and senior developers.

Key Project — Olotalk / Olochat AI Inc., Lead Developer

10/2025 – 05/2026

Project: Production AI-powered website chat and customer-support agent platform for Canadian SMBs, built as a pnpm monorepo and shipped as cloud-native SaaS.

Technologies: Python, Node.js/Express, TypeScript, React/Next.js, Qdrant, PostgreSQL, Redis, ClickHouse, Claude (LLM), RAG, Docker, Kubernetes, Linode, GitHub Actions

- Architected and built the platform end to end across a React/Next.js frontend and Node.js/Express backend in a pnpm monorepo.
- Designed a production RAG pipeline with DOM-aware segmentation, parent-child chunking, and contextual enrichment via Claude Haiku.
- Integrated Qdrant for vector search, PostgreSQL for primary storage, Redis for caching, session, and rate-limit state, and ClickHouse for analytics.
- Developed Python ingestion workers to crawl, parse, embed, and index website content into the vector store.
- Containerized and deployed services on Linode with Docker and Kubernetes, configuring autoscaling and zero-downtime rollouts.
- Built CI/CD pipelines with GitHub Actions and automated unit and integration tests across the monorepo.

BMO, Digital Core Cloud Center of Excellence, Toronto

05/2024 – 05/2025

Senior Full-Stack Developer

Project: Internal cloud-migration tool that parses and visually compares on-premises and cloud JSON configuration data during migration.

Technologies: Next.js, Node.js, AWS Lambda, AWS IAM (roles, policies, STS), DynamoDB, S3, GitHub Actions, Jenkins

- Built a Next.js/Node.js/Lambda module with a custom parser to compare on-premises and cloud JSON data.
- Implemented color-coded diff rendering for configuration comparison during migration.
- Designed secure access workflows with AWS IAM roles, policies, and STS to set permission boundaries for Lambda and S3.
- Integrated DynamoDB and S3 as cloud-native data stores for the comparison module.
- Applied Strategy and Observer patterns and a modular architecture to keep the parser and renderer extensible.
- Configured CI/CD pipelines (GitHub Actions, Jenkins) and wrote unit and integration tests for the comparison logic.

RBC, Enterprise Risk Management, Toronto

11/2022 – 05/2024

Senior Full-Stack Developer

Project: Interactive risk-exploration tool rendering draggable, interconnected charts so analysts can trace relationships across risk data.

Technologies: React, Redux, Node.js, Python/Django, Jest, Pytest, GitHub Actions; graph algorithms (Dijkstra, topological sort)

- Designed and implemented a React/Redux/Node.js and Python/Django tool rendering draggable charts interconnected via Dijkstra and topological sort.
- Integrated exploratory LLM-based tooling for data insights into the risk workflow.
- Built and maintained CI/CD pipelines (GitHub Actions) with automated Jest and Pytest test suites.
- Partnered with data scientists to map risk-management requirements to the visualization model.

HS GovTech Solutions Inc., Canada

02/2021 – 07/2022

Senior Full-Stack Developer

Project: myHD public-sector product, with backend logic migrated from a monolith to serverless AWS Lambda services.

Technologies: TypeScript, Node.js (Express), Vue.js, AWS Lambda, MS SQL Server, MongoDB, Jest, Mocha, GitHub Actions, Jenkins

- Maintained and extended the myHD product using TypeScript, Node.js, AWS Lambda, and Vue.js.
- Migrated monolithic backend logic to serverless architectures on AWS Lambda.
- Integrated MS SQL Server and MongoDB for backend data handling.
- Wrote automated unit and integration tests with Jest and Mocha and maintained CI/CD pipelines (GitHub Actions, Jenkins).

Funzo (Co-founder), Toronto

08/2020 – 08/2024

Co-Founder & Lead Instructor

Project: Co-founded a developer training program; built and taught the full-stack and cloud curriculum and led student project delivery.

Technologies: JavaScript, TypeScript, React/Next.js, Vue.js/Nuxt.js, Node.js, Golang, MongoDB, PostgreSQL, GCP (GKE, Cloud Run, Cloud Build, BigQuery), GitHub Actions, Jest, Mocha

- Designed and delivered curriculum in JavaScript, TypeScript, React/Next.js, Vue.js/Nuxt.js, Node.js, and Golang.
- Built a GCP module covering GKE, Cloud Run, Cloud Build, and BigQuery; students shipped containerized Node.js and Python services.
- Taught test-driven development with Jest and Mocha and CI/CD with GitHub Actions, and led cohorts through Agile sprints.

MobileLIVE, Toronto

10/2018 – 12/2020

Senior Full-Stack Developer

Project: Rogers and Fido digital properties (fido.ca, MyRogers) and CIBC/Simplii money-transfer features.

Technologies: Angular 7, AngularJS, NgRx, RxJS, TypeScript, Node.js, Karma, Jasmine, Jenkins, MongoDB, MySQL, PostgreSQL

- Led the migration of fido.ca from AngularJS to Angular 7 with NgRx, RxJS, and TypeScript.
- Built money-transfer features for CIBC/Simplii, including Global Money Transfer and transfers for international students.
- Maintained Karma/Jasmine test suites and Jenkins CI/CD across the migration.

Bell Media, Toronto

01/2018 – 09/2018

Senior Frontend Developer

Project: Vidi in-house video player (built on JW Player), used across 25 Bell brands.

Technologies: JavaScript, Vue.js, React, Angular, JW Player, Adobe Analytics, Git

- Built and maintained features for the Vidi video player deployed across 25 Bell brands.
- Integrated Adobe Analytics and ad components, and contributed Vue, React, and Angular components to the player ecosystem.

EARLIER EXPERIENCE

- **Ericsson, Toronto** (04/2017 – 12/2017) Led the frontend team building a Proximus (Belgium Telecom) dashboard with AngularJS, Node.js, and D3.js; ran React/Redux and TypeScript proof-of-concepts.
- **Isobar, Toronto** (11/2016 – 03/2017) Led a Disney project and an NDA data-visualization tool using Angular, Gulp, and Docker.
- **MobileLIVE, Toronto** (10/2015 – 10/2016) Built Rogers.com and Fido.ca features and an Angular 2/TypeScript/D3.js dashboard.
- **Blast Radius, Toronto** (04/2015 – 08/2015) Redesigned mw.theweathernetwork.com for cross-platform delivery.
- **Koda Media Group, Toronto** (01/2015 – 04/2015) Built the Air Canada Cargo site with Bootstrap and Gulp.js.
- **Next Pathway, Toronto** (02/2014 – 12/2014) Built a CMS proof-of-concept (Express, AngularJS) for Scotiabank and a customer-loyalty platform (Spring) for Aimia; integrated CKEditor and Video.js.
- **Critical Mass, Toronto** (05/2013 – 12/2013) Led the Citibank ThankYou.com redesign and UI architecture for Citibank and Nissan-Infiniti with Backbone.js, jQuery, and Jekyll.
- **TV Ontario, Toronto** (01/2013 – 04/2013) Redesigned ilc.org with Bootstrap and RESTful services.
- **Critical Mass, Toronto** (09/2012 – 12/2012) Built UI for Citibank mortgage accounts (Jekyll, JavaScript) and integrated Adobe Analytics.
- **CBC Television, Toronto** (09/2011 – 03/2012) Enhanced multimedia features on the CBC radio and video site.
- **Canadian Tire, Toronto** (11/2010 – 08/2011) Built client-side modules with a focus on XSS security and YSlow performance.
- **BabyRobot, Toronto** (05/2010 – 07/2010) Delivered the Keg and Nair sites under Agile and converted PHP code to Python/Django web services.
- **Starz Animation, Toronto** (03/2009 – 07/2009) Converted PHP code to Python/Django web services.
- **CGI Inc., Montréal** (08/2007 – 04/2008) Built web modules for the Deutsche Bank project (db.com) in Java JSP.
- **Forest Research Institute Rheinland-Pfalz, Germany** (02/2003 – 02/2006) Built three web applications (wald-rlp.de, oekosysansatz.de, adelsberg-lutzelhardt.de) with HTML, CSS, JavaScript, AJAX, and PHP.

EDUCATION & LANGUAGES

Languages: English, French, German

Education: M.S. in Electrical Engineering, University of Kaiserslautern-Landau, Germany (2006)